DO NOW — RECALL

Lesson 095 · CPU Pipelining: Overlap and Hazards

Name: _____ Class: _____ Date: _____

1. What is pipeline throughput?

2. What is a hazard?

3. Which CPU feature may reduce RAM accesses?

4. A D flip-flop has D=0 and Q=1 just before the active rising edge. What is Q after?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Rate of completed instructions.

2. Model answer

Dependency/resource/control issue that interrupts pipeline.

3. Model answer

Cache.

4. Model answer

Q becomes0, sampled at the active edge.

By the end, you can…

• Pipelining splits instruction processing into stages; several instructions can occupy different stages during the same clock cycle.

• With four independent instructions and three one-cycle stages, ideal completion is $3+4-1=6$ cycles; serial non-overlap takes $4\times3=12$.

• Data dependencies, branching and resource conflicts can stall or flush a pipeline. Throughput benefits need ideal assumptions, not automatic latency reductions.

Vocabulary — know these terms

pipeline — Stages overlapping different instructions.

throughput — Completed work per time.

data hazard — Dependency on unavailable result.

control hazard — Branch uncertainty interrupting fetch.

NEW LEARNING

Read, annotate and explain

Observe

Pipelining splits instruction processing into stages; several instructions can occupy different stages during the same clock cycle.

Explain

With four independent instructions and three one-cycle stages, ideal completion is $3+4-1=6$ cycles; serial non-overlap takes $4\times3=12$.

Evaluate

Data dependencies, branching and resource conflicts can stall or flush a pipeline. Throughput benefits need ideal assumptions, not automatic latency reductions.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Three instructions pass through a two-stage pipeline. Compute ideal completion cycles.

Explain the mechanism

2+3−1=4 cycles; serial baseline 2×3=6 cycles.

The second task starts stage one while first advances to stage two.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Show cycles for I1-I3 in a 3-stage F-D-E pipeline, and finish cycle of I3. [6]

Question 2

Compute ideal pipeline versus serial completion for 6 instructions and 3 stages. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why branch and data hazards can reduce actual improvement, distinguishing latency and throughput. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A pipeline makes each stage finish in less than one cycle by default.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.